Unit 1 Classwork — Real Numbers — Answer Key

Converting · Classification · Squares · Estimating Roots · Ordering

A. Converting Fractions ↔ Decimals

Problem 1 Score 3

Convert $\frac{3}{8}$ to a decimal.

Answer: 0.375

Problem 2 Score 3

Convert 0.45 to a fraction in simplest form.

Answer: $\frac{9}{20}$

Problem 3 Score 4

Convert $\frac{5}{6}$ to a decimal (round to the thousandths, use a repeating bar notation if needed).

Answer: $0.8\overline{3}$

Problem 4 Score 4

Convert $0.\overline{6}$ (repeating) to a fraction.

Answer: $\frac{2}{3}$

B. Classifying Numbers

Problem 1 Score 3

Is √16 rational or irrational?

Answer: rational

Problem 2 Score 3

Is √7 rational or irrational?

Answer: irrational

Problem 3 Score 3

Is π rational or irrational?

Answer: irrational

Problem 4 Score 4

Is 0.454545... (repeating) rational or irrational?

Answer: rational

Problem 5 Score 4

Is −5 a whole number, an integer, or both?

Answer: integer

C. Perfect Squares

Problem 1 Score 3

What is 12^2?

Answer: 144

Problem 2 Score 3

What is √225?

Answer: 15

D. Estimating Square Roots

Problem 1 Score 3

Between which two consecutive integers does $\sqrt{50}$ fall?

Answer: 7 and 8

Problem 2 Score 4

Estimate $\sqrt{50}$ to the nearest tenth.

Answer: 7.1

E. Ordering Rational & Irrational Numbers

Problem 1 Score 4

Which is greater: $\sqrt{2}$ or $\frac{3}{2}$?

Answer: $\frac{3}{2}$

Problem 2 Score 5

Order from least to greatest: $\sqrt{10}$, 3, $\frac{10}{3}$

Answer: 3, $\sqrt{10}$, $\frac{10}{3}$

Problem 3 Score 5

Which is greater: π or $\sqrt{10}$?

Answer: $\sqrt{10}$